On Market Making for a New Exchange: EUR/USD Designated Market Maker on Exchange A

Edouard Blanc Clément Dardare Julien Simeoni

Paris Dauphine University — MSc 203

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Contents

1 Introduction **2**
- 1.1 Motivation . . . 2
- 1.2 Objective . . . 2
- 1.3 Simulation Environment . . . 2
- 1.4 Evaluation . . . 3
- 1.5 Report Structure . . . 3

2 Order Book **4**
- 2.1 Data structure . . . 4
- 2.2 Submissions and cancellations . . . 4
- 2.3 Matching rule . . . 5
- 2.4 Fills as events . . . 5

3 Market Simulator for Venues B and C **6**

4 Market-Making Logic on Exchange A **7**

5 Applying Submissions and Cancellations **8**

6 Simulation Loop **9**

7 Inventory and P&L Tracking **10**

8 Phase 1 Backtest Report **11**

9 Phase 2 — Calibration from Trading History **12**

10 Phase 3 — Fallback Liquidity Under HFT Competition **13**

11 Conclusion **14**

1 Introduction

1.1 Motivation

A newly launched exchange — referred to throughout this report as exchange A — has mandated our firm as the sole Designated Market Maker for its EUR/USD spot book. Two established venues, B and C, remain the primary sites of price discovery, carrying on average 75% and 25% of the daily volume respectively. We receive market data from B with a 200 ms delay and from C with a 170 ms delay, while high-frequency participants reach the same data in 50 ms and can act on exchange A before our own view of fair value has refreshed. The mandate commits us to continuously quoting ten levels on each side of the book at a common 1 bp tick, with a starting capital K split evenly between EUR and USD and an asymmetric fee grid. A delta limit caps the EUR inventory we are willing to warehouse before externalising risk as taker flow on B and C.

Three constraints shape every design decision that follows. First, on the day exchange A goes live no historical fills or counterparty behaviour are observable, so the initial quoting policy cannot be learnt from data and must be derived from first-principles microstructure models. Second, the latency gap to HFTs exposes every resting quote to adverse selection, which the pricing model must compensate for rather than ignore. Third, because we cannot market-make on B or C, the delta limit turns inventory management into an active cross-venue hedging problem rather than a passive book-balancing exercise.

1.2 Objective

Our goal is to deliver an end-to-end market-making system that addresses the three constraints above and can be evaluated under the conditions described in the subject brief. The project is split into three phases matching the mandate timeline. Phase 1 bootstraps liquidity without any trading history, using a heuristic quoting policy derived from classical market-making theory. The fair mid is reconstructed from the two reference venues at their respective latencies, the bid-ask spread is sized to compensate for both inventory risk and adverse selection, and quotes are skewed in the direction of recent order-flow imbalance. A dedicated hedge router externalises excess EUR inventory on B and C whenever the delta limit is approached. Phase 2 re-estimates the Phase 1 parameters by maximum likelihood from a simulated month of fills. Phase 3 studies how our quoting policy should degrade from a price-competitive quoter to a reliable fallback liquidity provider once aggressive HFT market makers populate the inside of the book.

1.3 Simulation Environment

Because the exchange has no history at launch and we cannot run the strategy against real counterparties, the evaluation environment is a layered simulation built specifically for this project. A stochastic mid-price generator (GBM, GARCH with jumps, or Heston) produces the fair value underlying both reference venues. A venue simulator wraps that path into a realistic bid/ask and depth profile per venue, with a choice of five spread regimes (static, stochastic, adaptive, asymmetric, and skewed) reflecting the microstructure assumptions we want to stress. A limit order book with a matching engine and fill events sits at the centre of exchange A; a Poisson client-flow generator with truncated Pareto sizes delivers counterparty orders whose intensity decays exponentially in

the quoted half-spread; and a reporting layer reconstructs the full mark-to-market and realised P&L decomposition from the trade history.

1.4 Evaluation

Each phase is evaluated on the subject-mandated one-month simulated backtest, with the capital K reset to its initial split at the start of every phase. For Phase 1 we report realised and mark-to-market P&L, the percentile envelope of mark-to-market as a function of time since trade inception, top-ten trades by size, full-versus-partial fill rates at each of the ten quoting levels, and inventory trajectories against the delta limit. For Phase 2 we compare the heuristic and calibrated configurations on an independent month using the same diagnostics, so that any difference in performance is attributable to the updated parameters rather than to the market regime. For Phase 3 we measure both our own P&L under HFT competition and the quality of liquidity observed by exchange-A clients, distinguishing small from large trades and quantifying the continuity of two-sided quoting through volatility spikes, one-sided HFT retreats, and venue outages.

1.5 Report Structure

The remainder of the report follows the step-by-step structure of the subject. section 2 introduces the order-book data structure and matching rule; section 3 describes the venue simulator and the family of spread regimes supported; section 4 presents the quoting framework in full; section 5 covers submission and cancellation handling; section 6 documents the simulation loop that binds the modules together; section 7 details the inventory and P&L bookkeeping; section 8 gathers the Phase 1 backtest outputs; section 9 analyses the calibration results; section 10 examines the Phase 3 fallback behaviour; and section 11 summarises the findings and outlines the remaining modelling questions.

2 Order Book

The order book of exchange A must show a price and a quantity at ten levels on both sides at all times. This section describes the data structure we chose to hold that state, how submissions and cancellations flow through it, and the invariants of the matching engine. The design is exercised on a realistic EUR/USD seed (mid = 1.10, 1 bp tick, ten levels per side, 1M EUR per level).

2.1 Data structure

Two pressures shape the design. Matching requires a price-time-ordered view of the market-maker (MM) side at clearing time, yet the hot path is dominated by bulk requotes — up to ten cancels followed by ten posts per step. Keeping the book permanently sorted would pay on every mutation for a benefit consumed only once per step.

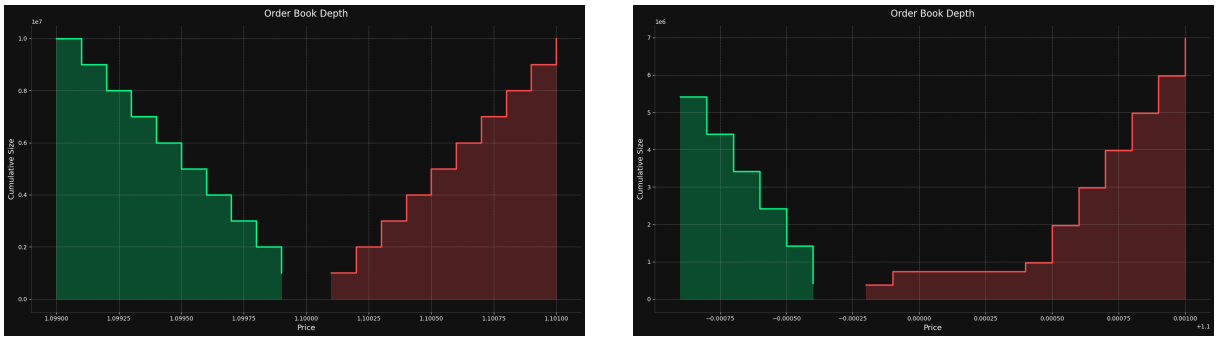
We therefore split orders by origin (MM versus client) into two hash-indexed stores and keep a price-ordered view of the MM side as a lazy cache, rebuilt only when the matcher runs. Cancels and posts touch only the hash stores; the MM side is sorted once per clearing call instead of being re-ordered on every mutation. Time priority within a price level is preserved by a monotonic sequence number assigned at insertion, so no timestamp arithmetic is ever needed.

Two alternatives were rejected. A Pandas-DataFrame primary store would copy the whole frame on every mutation, untenable at a millisecond cadence. A sorted container would keep the book ordered continuously, but the MM side is only read once per clearing call — paying that cost on every cancel and post is wasted work in a loop where requotes arrive as tight bursts. Pandas is kept only as a display layer: read-only views of the book and the match log are materialised on demand for notebook inspection.

Orders are immutable value objects. Partial fills decrement an internal remaining-size record rather than the resting order itself, and consumed levels are refilled by posting a fresh order for the gap. This append-only discipline keeps the match log a faithful ledger and decouples the ten-level contract from the engine: depth is a caller-supplied parameter, so the same code is reused unchanged in Phase 3 (section 10).

2.2 Submissions and cancellations

Mutations flow through four entry points, with a deliberate asymmetry between the two sides. On the MM side, cancellations and postings both arrive in bulk — a targeted list of ids or a wipe of the whole side on cancel, a full new ladder on post — and neither triggers the matcher, because MM postings are passive by construction and cannot cross. On the client side, each arrival is inserted individually and the matcher runs immediately, so every client order is reconciled against the current MM state before control returns. This mirrors the upstream flow: the Quoter requotes the whole ladder at once when its triggers fire, while clients arrive as a Poisson stream where each arrival is a potential match.



(a) Seeded book: one order per level, symmetric depth.

(b) After twenty marketable client orders: shallow levels consumed from the top.

Figure 1: Cumulative depth before and after a burst of client flow.

2.3 Matching rule

Three invariants govern every clearing call. Client orders only match against MM orders, never against each other: as the sole designated market maker for Phases 1 and 2 we are on one side of every trade, and the restriction removes an entire class of self-trading bugs. The fill price is always the passive MM price, never the incoming aggressive price — clients pay the spread we quote, which is the only source of gross revenue. And matching follows strict price-time priority with partial fills: the engine walks the sorted MM side, deducts the minimum of the two sizes, logs the match, and either retires the filled order or leaves the remnant resting. The residual shape in Figure 1 is a direct visual check of this rule: the top of the book is consumed first, deeper levels are untouched.

2.4 Fills as events

Every match is dispatched synchronously, as a fill event, to a single registered listener. In the full simulation the listener is the Quoter, which updates inventory, trade history, and the pending top-up queue inside the clearing call. The Quoter therefore never polls the book: the next quote computation already sees the post-fill inventory, which removes staleness bugs on steps that both fill and requote. Decoupling the book from its consumer also keeps the same engine usable in Phase 3, where a separate diagnostic listener records fills for the exchange-client-side liquidity analysis without touching the Phase 1 code path.

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